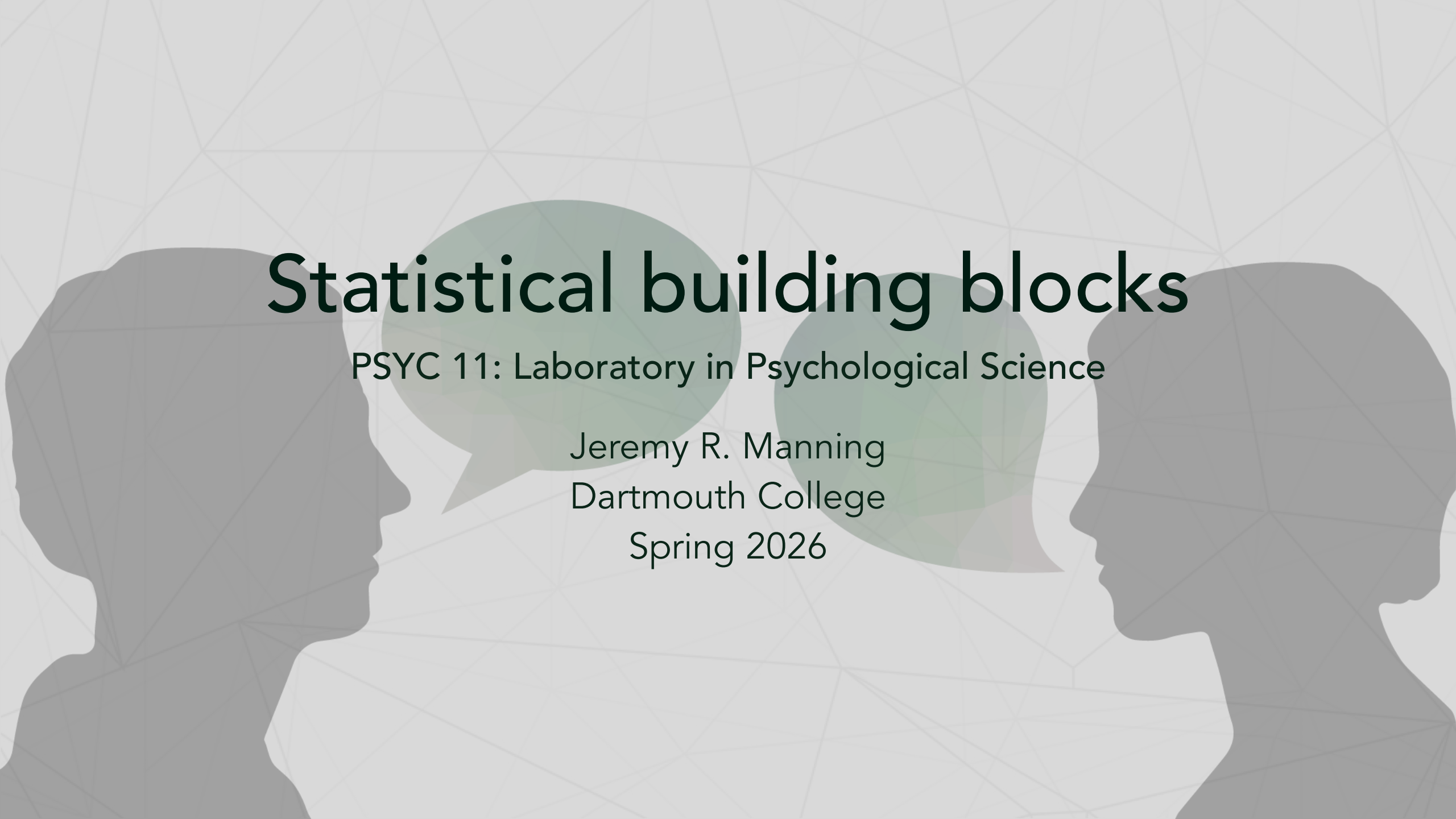




Statistical building blocks

PSYC 11: Laboratory in Psychological Science

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Spring 2026



The "intro to stats" view of stats

Choosing a test

You have several options to choose from:

- Want to compare the means of distributions? Use t-tests or ANOVAs
- Want to compare trends? Use correlations or regressions
- Etc.

Where do those tests come from?

Types of observations

Consider what sorts of values your observations can take on:

- Real numbers?
- Counts?
- Probabilities?
- Sets of numbers that sum to 1?

Where do those tests come from?

Key idea

Then we can ask: under different (typically very simple) assumptions about where the numbers came from, how likely would it be to see something like our actual data?

Where do those tests come from?

Example

Data: [-1.2, 2.04, 0.087, -0.1, ...]

How unexpected would these numbers be if:

- We thought the numbers came from a Normal distribution with mean = 0, var = 1
- We thought the numbers came from a Normal distribution with mean = 100, var = 1

Where do those tests come from?

Building tests from distributions

- To create the different tests you learn about in introductory stats courses, people have solved out the probabilities of observing different (sets of) values under different assumptions
- The p-value we get out tells us how unlikely it was that the observed data came from some "null" distribution

Making your own statistical tests

Distributions and parameters

- Different distributions can produce different types of draws-- Real numbers, counts, etc.
- Each distribution is typically controlled by one or more parameters-- mean/variance, probabilities, etc.

Probability distributions

Distribution name	Parameter(s)	What you get out	Example draws
Gaussian (Normal)	Mean, variance	Real numbers	-0.2, -10.923, 45.08, -6.4545
Bernoulli	Probability that $x = 1$	Results of "coin flips"	0, 1, 1, 0, 1, 0, 0, 0
Binomial	Number of observations, probability that each observation is 1	The number of observations where $x = 1$	10, 5, 38, 0, 267
Multinomial	Number of observations, probability that each <i>feature</i> in each observation is 1	The per-feature counts showing how many times each feature was 1	[3, 10, 2, 27], [46, 5, 4, 0]
Uniform	Start and end points (Real numbers)	A number between the start and end points	0.2, 0.7532, 0.00000123
Von Mises	Circular mean and concentration	Angles	3.6, 186, 240, 359.98

Making your own statistical tests

Recipe for a statistical test

1. Pick an appropriate distribution
2. Pick parameters that correspond to your "null hypothesis" (e.g., that the distribution has a mean of 0, that the values are equally likely, etc.)
3. Take a bunch of samples from your distribution
4. Compare the values of those samples to your actual data

Example: is a coin fair?

Setting up the problem

Suppose you observe some coin flips: $[0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, \dots]$

How could you figure out whether the coin is fair (e.g., explainable by 0 and 1 being equally likely)?

Example: is a coin fair?

Step 1: Choose a distribution

Suppose you observe some coin flips: $[0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, \dots]$

First, we need an appropriate distribution

Example: is a coin fair?

Step 2: Set up the null hypothesis

- Suppose you observe 12 coin flips: [0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0]
- For a fair coin, $p(1) = 0.5$
- We know that the number of "events" is 12 (i.e., the number of flips)
- Now we can ask: what's the probability of observing 3 or fewer 1s if the coin is fair?

Example: is a coin fair?

Step 3: Compute the p-value

- Suppose you observe 12 coin flips: [0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0]
- Parameters: $N = 12$, $p(x = 1) = 0.5$
- Now take a bunch of draws from the binomial distribution. Let's take 1,000,000 draws and ask: what proportion of those draws have a count less than or equal to 3?
- That's our p-value!

Demo

